

DEVI AHILYA VISHWAVIDYALAYA, INDORE**Institute of Engineering & Technology****B.E. First Year (IT-A and IT-B), Class Test-I****ACR1C2-Chemistry and Environmental Science**

Time: 70

Max Marks: 20

**Note: Attempt all questions. Answer must be brief and to the point.
Avoid use of additional Answer Sheet(s).**

Table 1: Analytical report of water sample A and sample B

SAMPLE A		SAMPLE B	
$\text{Ca}(\text{HCO}_3)_2$	0.001N	CaCl_2	0.01N
$\text{Mg}(\text{HCO}_3)_2$	0.002N	MgCl_2	0.01N
KCl	0.01N	HCl	36.5 mg/L
CaCl_2	55.5 ppm	NaAlO_2	82.0 ppm
MgCl_2	0.0475 g/L	NaCl	22.0 ppm

- Q.1 Define Hardness and their types. Calculate total, carbonate and non-carbonate hardness of water sample A.
- Q.2 Discuss basic principle of EDTA method. Determine normality of EDTA solution if 20 ml of water sample A is required 10.0 ml of EDTA solution using EBT as indicator.
- Q.3 Giving appropriate chemical reactions calculate the amount of lime (90% pure) and soda (85% pure) needed for softening of 1,00,000 liters of water sample B.
- Q.4 List chemical composition of Portland cement. Also explain chemical reactions taking place inside rotary kiln.

OR

List and explain (including their significance) various essential properties required in a Refractory
